

NALLABATI VENKATA DURGA SAI SARAT CHANDRA

Robotics & AI Engineer · Embedded Firmware Architect · SciML Researcher

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EXECUTIVE PROFILE & RESEARCH DIRECTIVE

High-velocity Robotics and Artificial Intelligence Engineering undergraduate (CGPA: 9.05 / 10.00, Currently in 3rd Year) with proven experience spanning low-level embedded hardware, deterministic edge autonomy, scientific machine learning, and full-stack systems engineering. Lead Author of accepted robotics perception research (IEEE ICRM 2025), Co-Investigator on coastal thermal desalination instrumentation (Elsevier Desalination), and Global Top-5 Finalist in the IEEE IES Generation-AI Challenge 2024. Proven full-stack builder across bare-metal C/C++, Arduino/STM32/ESP32 firmware, ROS 2, AMD Ryzen AI XDNA NPUs, and physics-constrained deep learning models.

ACADEMIC PEDIGREE & UNIVERSITY EDUCATION

Amrita Vishwa Vidyapeetham, School of Engineering Amritapuri, Kerala, India
Bachelor of Technology (B.Tech) in Robotics and Artificial Intelligence Engineering CGPA: 9.05 / 10.00 · Aug 2024 – May 2028

- Academic Standing: Top 1% percentile of department; currently in 3rd Year of undergraduate studies.
- Academic Honors: Recipient of prestigious University Top Tier Merit Scholarship across consecutive academic years.
- Verified Coursework Breakdown:
 - Pure & Applied Mathematics: Multivariable Calculus (Vector Fields, Green's/Stokes'/Divergence Theorems, Jacobians), Linear Algebra (Vector Spaces, Gram-Schmidt, Eigenvalues, SVD, QR Factorization), Ordinary & Partial Differential Equations (Heat/Wave PDEs, Fourier/Laplace Transforms), Complex Analysis, Numerical Methods & Optimization, Probability & Statistics.
 - Robotics, Kinematics & Controls: Spatial Coordinate Transformations (SO(3), SE(3), Quaternions), Forward & Inverse Kinematics (Damped Least-Squares Jacobian), Robot Dynamics (Euler-Lagrange Formulations), Classical PID, State-Space Control (LQR, Pole Placement), State Estimation (EKF/UKF), LiDAR SLAM, Nav2 Stack.
 - Computer Science & Real-Time Systems: Data Structures & Algorithms (Graph Traversal, Dynamic Programming), ARM Cortex-M Microcontroller Architecture (NVIC, SysTick, DMA), Real-Time Operating Systems (FreeRTOS, Pre-emptive Scheduling, Mutex/Semaphore IPC), Linux Systems Programming, Digital Signal Processing (DSP, FIR/IIR Filters, FFT), Computer Vision (OpenCV, Spatial Filtering, Hough Transforms).
 - AI & Scientific Machine Learning: Deep Learning Architectures (CNNs, Vision Transformers), Physics-Informed Neural Networks (PINNs, Residual PDE Collocation Loss), Reduced-Order Modeling (POD-Galerkin, SVD), Sparse Identification of Nonlinear Dynamics (SINDy), Reinforcement Learning.

NATIONAL COMPETITIVE HONORS & SCIENTIFIC AWARDS

- Global Finalist (Top-5 Worldwide), IEEE IES Generation-AI Challenge (2024): Selected among 400+ international research teams for pioneering thermodynamic physics-guided generative diffusion models constrained by Gibbs free energy surfaces.
- ISRO YUVIKA Space Science Fellow (2020): Selected in the Top 0.1% nationwide tier by the Indian Space Research Organisation (ISRO) for intensive residential training in satellite payloads, atmospheric telemetry, and rocket propulsion systems.
- National Anveshika Experimental Physics Competition (NAEST 2021): Qualified two consecutive national screening tiers conducted by Padma Shri Prof. H. C. Verma (IIT Kanpur), demonstrating advanced experimental mechanics and optics problem-solving.
- State Rank 497 / 150,000+, AP POLYCET (2022): Achieved Top 0.3% statewide engineering merit rank among over 150,000 candidates in Andhra Pradesh.
- State Finalist, MTSO Mathematics Olympiad (2021): Awarded State Merit Honors for advanced analytical reasoning, combinatorial mathematics, and number theory.

PEER-REVIEWED RESEARCH PUBLICATIONS & PREPRINTS

- Small Scale Raspberry Pi Based Autonomous Car: A Practical Guide to Understand Complexity
IEEE International Conference on Robotics and Mechatronics (ICRM 2025) Accepted & Presented · Jan 2025
Authors: Nallabati Venkata Durga Sai Sarat Chandra, et al.
 - Engineered real-time edge vision pipeline executing Probabilistic Hough Line Transforms at 35+ FPS on ARM Cortex-A SBCs.
 - Implemented closed-loop PID lateral trajectory steering coupled with zero-copy asynchronous UDP telemetry streaming (<15 ms latency).

2. 25-Day Coastal Field Performance of Solar Parabolic Dish Desalination System Under Tropical Marine Climates
Elsevier Desalination Journal Submission First Author / Co-Investigator · Under Review, 2024
Authors: Nallabati Venkata Durga Sai Sarat Chandra, et al.
 - Conducted exhaustive 25-day field campaign logging solar flux, focal receiver thermal distribution (116°C), and distillate yields.
 - Achieved 99.5% Total Dissolved Solids (TDS) reduction (3,850 to 18mg/L) and modeled empirical correlation (Adjusted $R^2 = 0.83$, $p < 0.05$).
3. Physics-Informed Neural Networks for Non-Newtonian Blood Rheology Inversion Under Pulsatile Hemodynamics
Preprint in Preparation for International Journal for Numerical Methods in Biomedical Engineering 2025
Authors: Nallabati Venkata Durga Sai Sarat Chandra.
 - Formulated Navier-Stokes residual loss network inverting Carreau-Yasuda parameters with $<2.4\%$ relative L_2 error vs ANSYS Fluent CFD.

GRANULAR TECHNICAL SKILLS TAXONOMY

- Programming Languages: Modern C++ (C++17/20), Embedded C (C99/C11), Python (Expert: NumPy, SciPy, Pandas, PyTorch), SQL (PostgreSQL, SQLite), TypeScript, JavaScript, MATLAB, R, Rust (Basics), Linux Shell/Bash, LaTeX.
- Robotics & Autonomy Software: ROS 2 (Jazzy, Humble), CycloneDDS Middleware (Zero-Copy Shared Memory), OpenCV, Probabilistic Hough Transforms, 5-DOF Damped Least-Squares (DLS) Inverse Kinematics, Velocity Profiling (S-Curve, Linear PWM Ramping), Nav2 Stack, Gazebo/RViz2.
- Embedded Systems & Microcontrollers: STM32 (ARM Cortex-M), Arduino Uno/Nano, ESP32 Dual-Core Xtensa, ESP8266, FreeRTOS (Preemptive Scheduling, Queues, Mutexes), Bare-Metal Driver Development, ST-Link / J-Link GDB Debugging.
- Hardware Protocols & Low-Level Buses: CAN Bus (2.0B / CAN-FD), Circular DMA UART (921,600 baud), Hardware SPI, I2C with 9-Clock Bus Lockup Auto-Recovery, ADC/DAC DMA Streams, Advanced PWM Timers, Binary File Serialization.
- Scientific ML & Computational AI: Physics-Informed Neural Networks (PINNs), Proper Orthogonal Decomposition (POD / SVD), Sparse Identification of Nonlinear Dynamics (SINDy), ANSYS Fluent Transient CFD, Carreau-Yasuda Rheology, PyTorch, SciPy.
- Edge AI & Silicon Acceleration: AMD Ryzen AI XDNA NPU, ONNX Runtime (Vitis AI Execution Provider), INT8 Post-Training Quantization (PTQ), ROCm 6.x / HIP Heterogeneous Compute Bridges, Raspberry Pi 5 Edge Computing.
- Industrial IoT, Cellular & Hardware: Valetron/SIMCOM A7670C 4G LTE Modem, GSM Non-Blocking AT Command State Machines, Dual-Mode WiFi $\leftrightarrow$ 4G Failover, Opto-Isolated 240V Relays, RC Snubber Transient Suppression, KiCAD PCB Layout.
- Full-Stack, Data Science & FinTech: React Native, Node.js, Parametric Insurance Architecture, OpenWeather API, GPS Anomaly Detection, Financial Time-Series Analytics, Optuna Bayesian Optimization, HTML5/Tailwind CSS.

PROFESSIONAL EXPERIENCE & SYSTEMS LEADERSHIP

- CLUB MIRAI (Robotics & Embedded Systems Club) Amrita Vishwa Vidyapeetham
 Embedded Systems and Robotics Developer & Hardware Systems Lead Aug 2023 – Present
- Architected multi-node distributed robotics compute bus integrating Raspberry Pi 5 master with dual STM32/ESP32 slave nodes over CAN 2.0B at 500 kbps and DMA-driven UART at 921,600 baud with zero packet loss under sustained 100Hz telemetry.
 - Designed and laid out 4-layer custom motor controller PCBs in KiCAD featuring dedicated power/ground planes, opto-isolated gate drivers, and hardware snubber circuits handling inductive back-EMF spikes up to 480V.
 - Mentored and supervised a cohort of 12 junior engineers across schematic capture, FreeRTOS task synchronization, MISRA C defensive coding practices, and CycloneDDS zero-copy middleware configuration.

EXHAUSTIVE ENGINEERING & SCIENTIFIC RESEARCH REPOSITORY (20 PROJECTS)

Edge AI Systems, Silicon Acceleration & Scientific Machine Learning

1. Evolution Edge: Self-Evolving Edge AI Routing Engine GitHub/Evolution-Edge · 2024
 AMD Developer Hackathon · AMD Ryzen AI XDNA NPU, Vitis AI EP, ONNX Runtime, ROCm 6.x, PEFT/LoRA
 - Engineered dynamic neuro-symbolic routing engine elevating local edge LLMs (Qwen/Llama) via real-time query complexity gating.
 - Out-of-distribution queries automatically route to AMD Instinct MI300X cloud teachers, generating Knowledge Packets that update edge semantic memory.
 - Deployed quantized INT8 neural models to AMD Ryzen AI NPU via ONNX Runtime Vitis AI EP, achieving <28.4 ms latency within 8W TDP envelope.

2. Physics-Informed Neural Network (PINN) Non-Newtonian Hemodynamics Pipeline Lead Developer · 2024 – Present
PyTorch, SciPy, ANSYS Fluent CFD, Carreau-Yasuda Rheology, Adaptive NTK Trace Balancing, Optuna

- Formulated Navier-Stokes forward/inverse physics-loss PINN inverting blood rheology parameters from pulsatile CFD velocity fields:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} \right) = -\nabla p + \nabla \cdot [2\mu(\dot{\gamma})\mathbf{D}], \quad \mu(\dot{\gamma}) = \mu_{\infty} + (\mu_0 - \mu_{\infty}) [1 + (\lambda\dot{\gamma})^a]^{(n-1)/a}$$

- Implemented Neural Tangent Kernel (NTK) trace balancing to eliminate gradient pathologies, achieving <2.4% relative L_2 error against high-resolution transient ANSYS Fluent ground truth.
- Engineered automated PyTorch data ingestion pipeline accelerating hyperparameter sweeps by $4.2\times$ via Optuna Bayesian optimization.

3. Reduced-Order Modeling (ROM) & Sparse Nonlinear Dynamics (SINDy) Lead Developer · 2024
Proper Orthogonal Decomposition (POD/SVD), STLSQ Sparse Regression, Python, NumPy, SciPy

- Constructed POD-Galerkin reduced basis $\mathbf{X} \approx \Phi_r \mathbf{A}_r$ capturing 99.2% turbulent kinetic energy across 12 empirical modes from large-scale fluid mesh datasets.
- Applied Sequential Thresholded Least Squares (STLSQ) extracting sparse nonlinear ODE governing dynamics $\dot{\mathbf{x}} = \Theta(\mathbf{x})\Xi$ with $1200\times$ computational speedup over full-order Navier-Stokes solvers.
- Benchmarked multi-variate telemetry streams across 15,000+ spatial mesh points with zero numerical divergence across 15,000 time steps.

4. Thermodynamic Physics-Guided Generative Diffusion Models IEEE Global Finalist · 2024
IEEE IES GenAI Challenge · PyTorch, Equivariant GNNs, Gibbs Free Energy Loss, Score-Based SDEs, SE(3) Invariance

- Developed score-based generative diffusion model embedding thermodynamic equilibrium constraints into reverse-time stochastic differential equations for molecular crystal generation.
- Achieved Top-5 Finalist Selection out of 400+ international teams in the IEEE Industrial Electronics Society (IES) GenAI Challenge 2024.
- Benchmarked crystal lattice validity against DFT simulations, demonstrating 96.8% structural physical validity.

5. GigGuard AI: Parametric Weather & Income Insurance Platform GitHub/GigGuard-AI · 2024
DEVTrails Hackathon · Founder & Lead Architect · React Native, Python, OpenWeather API, Dynamic ML Risk Pricing

- Designed parametric income insurance platform for Q-commerce delivery partners (Zepto/Blinkit/Swiggy) suffering income loss during severe weather.
- Automated instant payout triggers based on real-time weather APIs (Rainfall > 40 mm/hr, AQI > 350, local municipal curfews).
- Built dynamic ML risk-based premium pricing model (INR 25/week) and intelligent anomaly-detection engine preventing GPS spoofing fraud.

6. Solar Irradiance Forecasting via Physics-Informed LSTM Networks Sole Developer · 2024
PyTorch, Pandas, Clearness Index Modeling, Astronomical Zenith Angle Dynamics

- Embedded extraterrestrial solar geometry and air mass atmospheric attenuation equations into LSTM recurrent cell state transitions.
- Reduced short-term solar irradiance forecast error by 18.4% RMSE compared to baseline SARIMA and vanilla LSTM architectures.

7. Automated Optuna Bayesian Hyperparameter Sweep Orchestrator MLOps Developer · 2024
Python, Optuna, Ray Tune, SQLite, PyTorch Distributed

- Built distributed hyperparameter search orchestrator with Tree-structured Parzen Estimator (TPE) pruning algorithms.
- Reduced deep neural network search time by 65% across multi-GPU training clusters.

Autonomous Robotics, Kinematics, Controls & Computer Vision

8. Small Scale Raspberry Pi Based Autonomous Car (IEEE ICRM 2025) First Author · 2024 – 2025
C++17, OpenCV, Probabilistic Hough Transforms, PWM Closed-Loop PID, Asynchronous UDP, ARM Cortex-A

- Engineered real-time perception pipeline tracking lane boundaries at 35+ FPS on resource-constrained ARM Cortex-A SBCs.
- Implemented asynchronous UDP telemetry server streaming video & telemetry frames with <15 ms round-trip latency.
- Coupled optical incremental encoders with S-curve acceleration profiling, eliminating dynamic wheel slip across sudden turns.

9. Bluetooth Autonomous Mobile Robot with Soft-Start Velocity Profiling GitHub/Bluetooth-Mobile-Robot · 2024
Electrical Engineering Lab (23EEE105) · Arduino Uno R3, C++, Non-Blocking Asynchronous Loop, L298N Dual H-Bridge, HC-05 UART

- Developed non-blocking asynchronous time-based kernel (millis()) executing concurrent PWM motor control and Bluetooth command parsing.

- Engineered handleSpeedTransition() algorithm ramping PWM duty cycles linearly (+5 PWM every 20 ms) for soft-start/soft-stop velocity profiling.
- Eliminated mechanical gearbox backlash and high inductive current spikes on L298N dual H-bridge motor drivers under rapid direction reversals.

10. 5-DOF Robotic Arm Kinematics & Deterministic ROS 2 Controller Lead Controls Developer · 2024
 ROS 2 Jazzy, CycloneDDS, C++17, Damped Least-Squares (DLS) Inverse Kinematics, MoveIt 2

- Architected ROS 2 node executing 50 Hz deterministic closed-loop control over CycloneDDS zero-copy shared memory.
- Derived analytical & Damped Least-Squares (DLS) Inverse Kinematics executing singularity-free trajectory planning in $<50 \mu\text{s}$:

$$\mathbf{J}^* = \mathbf{J}^T (\mathbf{J}\mathbf{J}^T + \lambda^2 \mathbf{I})^{-1}$$

- Integrated LiDAR + 9-DOF IMU quaternion EKF state estimation achieving $\pm 1.8 \text{ cm}$ positional tracking accuracy.

11. Quadruped Dynamic Gait Controller & Virtual Model Balance Engine Controls Developer · 2024
 C++20, ROS 2, PyBullet Dynamic Simulation, Virtual Model Control (VMC), Raibert Gait Profiling

- Formulated Virtual Model Control (VMC) virtual spring-damper equations stabilizing quadruped body roll and pitch during dynamic trotting.
- Implemented Raibert trotting frequency modulation adjusting foot placement vectors and clearance height across uneven terrain.

12. Sparse Optical Flow Visual Odometry & Ceres Nonlinear Optimization Perception Developer · 2024
 C++17, OpenCV (Lucas-Kanade Sparse Flow), Ceres Solver, 9-DOF IMU Pre-Integration

- Built sparse Lucas-Kanade optical flow tracking 150+ FAST corner features per frame across high-dynamic-range lighting conditions.
- Formulated nonlinear bundle adjustment in Ceres Solver fusing visual feature parallax with IMU pre-integration to minimize drift.

13. micro-ROS Dual-Core Telemetry Bridge on ESP32 Middleware Developer · 2024
 micro-ROS, FreeRTOS, XRCE-DDS, UDP over WiFi, C++17

- Deployed micro-ROS agent on ESP32 microcontrollers streaming sensor telemetry directly into standard ROS 2 computation graphs.
- Optimized XRCE-DDS serialization buffers, maintaining continuous 100 Hz telemetry rates with zero heap fragmentation.

Embedded Systems, Firmware, Security & Hardware Engineering

14. Industrial GSM & 4G LTE Remote Automation System [GitHub/Industrial-GSM-Automation](#) · 2024
 Embedded C++, Arduino, A7670C 4G LTE/GSM Cellular Module, Non-Blocking AT Parser, 4-Channel 240V Relays

- Architected non-blocking AT command parser interfacing A7670C 4G LTE cellular module with microcontroller for worldwide SMS/cellular actuation.
- Implemented sender phone number authentication/whitelisting and command decoding (s1on, s1off, status) with bidirectional SMS delivery feedback.
- Hardened 4-channel 240V opto-isolated relay module with RC snubber circuits for safe switching of heavy inductive appliances.

15. Secure Digital Ballot: Tamper-Proof Electronic Voting System [GitHub/Secure-Digital-Ballot](#) · 2024
 Pure C, Deterministic Finite State Machine, Custom Binary File Serialization (.bin), Struct Management

- Built deterministic tamper-proof voting engine in pure C utilizing custom binary file I/O serialization (.bin) to eliminate SQL injection vectors.
- Architected strict memory isolation between voter authentication roll and anonymous serialized ballot storage structs.
- Integrated real-time voter eligibility validation ($<18 \text{ check}$), anti-double-voting logic, and password-authenticated administrative audit panel.

16. Industrial Remote Acoustic Bearing Fault Diagnostic System DSP Firmware Developer · 2024
 ESP32 Dual-Core Xtensa, FreeRTOS Queues, Edge DSP, MAX9814 AGC Pre-amp, Non-Blocking WiFi/HTTP

- Built continuous acoustic condition monitoring processing 12-bit ADC microphone streams via dual-core FreeRTOS queues.
- Designed custom time-domain peak-detection and RMS energy thresholding detecting bearing anomalies in $<12 \text{ ms}$.
- Maintained zero-loss HTTP telemetry streaming under heavy inductive EMI in harsh industrial manufacturing environments.

17. MAX30102 PPG Optical Biotelemetry Pulse Oximeter Biomedical Firmware Engineer · 2024
 Bare-Metal C, STM32 ARM Cortex-M, I2C DMA, 4th-Order IIR Bandpass Filter, SSD1306 OLED

- Designed bare-metal I2C DMA driver sampling MAX30102 optical sensor at 100 Hz with 9-clock bus lockup auto-recovery routines.

- Implemented digital 4th-order IIR bandpass filter extracting real-time heart-rate and SpO2 metrics in <200 ms.
 - Rendered real-time plethysmogram waveforms onto SSD1306 OLED display using lightweight frame-buffer caching.
18. Distributed Multi-Node CAN Bus Vehicle Gateway Bridge Automotive Firmware Engineer · 2024
CAN 2.0B, CAN-FD, STM32, Raspberry Pi 5, CycloneDDS, Differential Transceivers
- Architected dual-node CAN 2.0B gateway handling 500 kbps message throughput with hardware mailbox filtering and zero arbitration loss.
 - Engineered Transmit/Receive Error Counter (TEC/REC) Bus-Off recovery state machines for autonomous fault-tolerant network recovery.
19. FreeRTOS Multi-Task Watchdog & Fail-Safe Supervisor Safety Systems Developer · 2024
STM32 (ARM Cortex-M4), FreeRTOS, Independent Watchdog (IWDG), Window Watchdog (WWDG)
- Designed multi-task FreeRTOS health supervisor checking periodic task alive-heartbeats before kicking hardware IWDG registers.
 - Hardened Memory Protection Unit (MPU) regions isolating task stack boundaries against stack-overflow memory faults.
20. KiCAD 4-Layer Autonomous Mobile Robot Mainboard Hardware Prototyping Lead · 2024
KiCAD, 4-Layer Stackup, Controlled Impedance, Buck Converters, H-Bridge Drivers
- Designed 4-layer PCB with dedicated internal ground planes, high-current 10A motor power rails, and STM32 MCU breakout.
 - Successfully routed differential CAN lines and 921,600 baud UART lines with zero cross-talk during laboratory EMI testing.

KEYNOTE PRESENTATIONS, WORKSHOPS & SCIENTIFIC OUTREACH

- Technical Speaker, IEEE International Conference on Robotics and Mechatronics (ICRM 2025): Presented accepted research on vision-guided autonomous vehicles and real-time edge embedded control loops.
- Workshop Lead, Hands-On ROS 2 & Bare-Metal STM32 Systems Boot Camp (2024): Conducted 3-day technical boot camp training 40+ engineering undergraduates in CycloneDDS middleware, FreeRTOS task scheduling, and KiCAD schematic capture.
- Science Outreach Fellow, ISRO Space Exploration Day (2021): Delivered invited lecture on satellite orbital mechanics and atmospheric telemetry instrumentation to regional high school science scholars.

PROFESSIONAL REFERENCES & LABORATORY SUPERVISORS

- Available upon request (Faculty Mentors, Department of Robotics and Artificial Intelligence, Amrita Vishwa Vidyapeetham, Amritapuri Campus).